Johanna Jacob

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CAREER OBJECTIVES

My career objective is to inspire and prepare the next generation of computing professionals through engaging, student-centered teaching. I aim to give students a strong foundation in computer science and cybersecurity by emphasizing hands-on learning, critical thinking, and real-world problem-solving, while mentoring them through research, competitions, and career pathways. My research in security and privacy, cybersecurity education, security for vulnerable populations, and AI sustainability directly informs my teaching, allowing me to bring current challenges into the classroom and involve undergraduates in meaningful scholarly work.

RESEARCH INTERESTS

- Cybersecurity Education, Training, and Workforce Development
- Broadening Participation in Computer Science
- Privacy and Security

EDUCATION

The University of Texas at San Antonio **Aug 2019 - Dec 2024**
Doctor of Philosophy in Computer Science *San Antonio, Texas*

- Advisor: Dr. Greg White, Director, Center for Infrastructure Assurance and Security, UTSA
- Dissertation: Examining the Divide: Measuring the Inequitable Pentration and Access to K-12 Cybersecurity Education
- GPA: 3.92

University of Houston - Clear Lake **Aug 2016 - Dec 2018**
Master of Science in Computer Science *Houston, Texas*

- Advisor: Dr. Andrew Yang, Associate Professor, CS/CIS
- Thesis: Towards a Holistic Interdisciplinary Cybersecurity
- GPA: 3.6/4.00

Loyola Institute of Technology **Sep 2009 – May 2013**
Bachelor of Science in Computer Science *Chennai, India*

PROFESSIONAL EXPERIENCE

Department of Computer Science, Trinity University **Aug 2025 - Present**
Tenure-Track Assistant Professor *San Antonio, Texas*

Teach undergraduate core and elective courses in Computer Science, and Cybersecurity. Mentor students in research and advise on academic and career pathways. Lead a research agenda . Advise the Trinity Cyber Club and engage students in hackathons and capture-the-flag competitions. Pursue professional development in cybersecurity and connect with industry partners to bring real-world learning into the classroom. Contribute to curriculum development and department and university service. Courses taught include:

- CSCI 1320 - Principles of Computer Science I - Fall 2026
- CSCI 2195 - Competitive Programming in Cybersecurity - Fall 2026
- CSCI 3190 - Directed Study (2 sections) - Fall 2026
- CSCI 3411 - Information Assurance and Security - Fall 2026

- CSCI 3191 - Computer Science Experiential Learning - Summer 2026
- CSCI 1320 - Principles of Computer Science I - Fall 2025
- CSCI 3199 - Salon d'informatique - Fall 2025
- CSCI 3354 - Big Data and Machine Learning - Fall 2025
- CSCI 3391 - Computer Science Experiential Learning - Fall 2025

Department of Computer Science, Trinity University **Aug 2024 - Aug 2025**
Visiting Assistant Professor *San Antonio, Texas*

Taught three undergraduate courses per semester, including Database Systems and Principles of Computer Science I, using student-centered teaching methods and hands-on projects to enhance learning and engagement. Provided academic support and mentorship through regular office hours and guidance on term projects, fostering an inclusive learning environment that encourages student growth and success.

- CSCI 3091 - Computer Science Experiential Learning - Summer 2025
- CSCI 3190 - Directed Study - Summer 2025
- CSCI 2320 - Principles of Data Abstraction (2 sections) - Spring 2025
- CSCI 3394 - Seminar: Cyber Operations - Spring 2025
- CSCI 1320 - Principles of Computer Science I (2 sections) - Fall 2024
- CSCI 3343 - Database Systems (2 sections) - Fall 2024

UTSA Roadrunner Assistantship Academy **Aug 2022 - Aug 2023**
Student Ambassador and Advocate *San Antonio, Texas*

Completed the Roadrunner Assistantship Academy (RAA), a year-long intensive program that helps aspiring student educators refine their pedagogical techniques in computing. Collaborated with the Academic Innovation team to analyze contemporary teaching techniques, apply generative AI tools in teaching and learning, evaluate emerging educational technologies, and train other graduate teaching assistants and graders.

Department of Computer Science, UTSA **Aug 2019 - May 2023**
Graduate Teaching Assistant *San Antonio, Texas*

Assisted faculty with classroom instruction, recitations, grading, record keeping, and other projects; corresponded with students and provided constructive feedback at the discretion of the instructor. Courses supported include:

- CS1173 - Data Analytics and Visualization using R Programming - Fall 2019 (Recitations)
- CS1173 - Data Analytics and Visualization using MATLAB - Spring 2020 (Recitations)
- CS2073 - Computer Programming with Engineering Application - Spring 2020, Fall 2021
- CS3733 - Operating Systems - Summer/Fall 2021
- CS4673 - Cyber Operations - Fall 2020/Spring 2022
- CS3113 - Principles of Cyber Security (Undergraduate) - Spring 2021/Fall 2021
- CS5323 - Principles of Cyber Security (Graduate) - Spring 2022/Fall 2022
- CS5343 - Secure Systems and Software - Fall 2022
- CS5733 - Privacy Enhancing Techniques - Spring 2023
- CS6343 - Cyber Risk Management - Spring 2023

Soaring Hawks' Girls Code Academy, UHCL **Aug 2017 - Aug 2019**
Primary Instructor *Houston, Texas*

Served as a primary instructor for 75 middle school girls (grades 6–8) from the foster community. Designed and delivered the camp curriculum, providing instruction in computer coding while developing students' problem-solving and analytical thinking to encourage interest in STEM careers. Trained other staff and instructors on course material and software accommodations.

KidsU Summer Camp, University of Houston **Jun - Jul 2017, 2018, 2019**
Primary Instructor *Houston, Texas*

Served as a primary instructor for three summers at KidsU, the flagship technology camp of the University of Houston, for middle and high school students in Clear Creek ISD. Designed and delivered curriculum for courses in Python, Web Design and Development using HTML/CSS, Game Design and Development, and Robotics.

UHCL Writing Center **Jan - Dec 2017**
In-house and Embedded Writing Center Consultant *Houston, Texas*

Assisted student and faculty writers from all disciplines at every stage of the writing process through face-to-face and distance sessions, including theses and dissertations. Provided feedback on writing in general and on individual drafts, and collaborated with faculty partners to extend Writing Center services into undergraduate writing courses through course-embedded consulting.

RESEARCH EXPERIENCE

Center for Infrastructure Assurance and Security, UTSA **May 2023 - Aug 2024**
Graduate Research Assistant *San Antonio, Texas*

Served as Research Assistant on an NSF-funded project, the Whole Community Cybersecurity Initiative, led by CIAS (Center for Infrastructure Assurance and Security), across pilot communities in San Angelo and Eden. Applied a comprehensive approach to cybersecurity spanning government agencies, businesses, educational institutions, and the general public. Planned, implemented, and assessed cybersecurity civil-defense capabilities tailored to the unique needs of the K–12 community. Developed impactful training modules and workshops, trained teachers and school staff on cybersecurity fundamentals, and guided instructors in the use of cybersecurity instructional resources. Assisted with IRB acquisition, collected and analyzed feedback, managed data collection, and contributed to collaborative project meetings, conference presentations, and a peer-reviewed publication.

NSF CyberCorps Project, UHCL **Aug 2017 - Aug 2019**
Graduate Research Assistant *Houston, Texas*

Played a key role in implementing a modernization strategy for a conventional Computer Science degree program, focusing on integrating cybersecurity knowledge and skills into a traditional four-year curriculum. Conducted rigorous self-study, performed gap analyses, and contributed to curricular development. Responsibilities included conducting extensive literature reviews, aligning course materials with industry frameworks such as NIST SP800-181, collecting and analyzing data using mixed methods research, developing and evaluating cybersecurity courseware and lab modules, generating progress reports, publishing papers in relevant venues, and contributing to overall project advancement. Applied learning analytics to assess student success rates within the modernized Computer Science degree program.

PUBLICATIONS

- **Jacob, Johanna** and Greg White. "Does Rural Matter: A Case Study on Under-represented Populations in a National Cybersecurity Competition". In preparation for *ACM Research on Equity and Sustained Participation in Engineering, Computing, and Technology (RESPECT)*, 2027 [**In Progress**]
- **Jacob, Johanna**, Scarlet Clouse, and Greg White. "Dissecting Disparities in K-12 Cybersecurity: Identifying Trends, Patterns and Participation Gaps in National Cybersecurity Competitions". In preparation for *Journal of Cybersecurity Education, Research, and Practice* [**In Progress**]
- **Jacob, Johanna** and Gideon Stanley. "Beyond PCI-DSS: Cybersecurity Protocol Adherence Among Small Hospitality Businesses". In preparation for *Symposium on Usable Privacy and Security* [**In Progress**]
- Florence Martin and **Jacob, Johanna**. "Cyberattacks in K-12 Education: A Case Study Applying the Diamond Model and Recommendations". In preparation for [**In Progress**]

- Owen Story and **Jacob, Johanna**. "Characterizing SMS Phishing Tactics, Patterns, and Detection Evasion Techniques". *IEEE MetroCon*, 2026 [**Under Review**]
- Nitya Donthireddy, Iniya Gopal, Raphael Fischer, and **Jacob, Johanna**. "Testing the Accuracy of AI Energy Consumption Tools". *IEEE MetroCon*, 2026 [**Under Review**]
- Eduardo Davalos Anaya, Yike Zhang, Katherine A. Troyer, Aaniyah Alexander, **Johanna Jacob**, and Britton Horn. "Games Before Class: Student Perspectives on Play, Belonging, and Learning Readiness in Higher Education."*In Proceedings of the International Conference of the Learning Sciences (ICLS), 2026*
- Greg White, Natalie Sjelin, Larry Sjelin, and **Jacob, Johanna**. "Educational Games to Teach Fundamental Principles of Cybersecurity". In *Proceedings of the 59th Hawaii International Conference on System Sciences*, 2026.
- **Jacob, Johanna**. "Examining the Divide: Measuring the Inequitable Penetration and Access to K-12 Cybersecurity Education". Doctoral Dissertation, The University of Texas at San Antonio, 2024.
- **Jacob, Johanna** and Greg White. "Is a Significant Demographic Left Out of the Equation? An Overview of Possible Inequitable Access to Cybersecurity Educational Programs in the United States". In *Proceedings of the 55th Hawaii International Conference on System Sciences*, 2022.
- **Jacob, Johanna**, Michelle Peters, and T. Andrew Yang. "Interdisciplinary Cybersecurity: Rethinking the Approach and the Process". In *National Cyber Summit (NCS) Research Track*, pp. 61–74. Springer International Publishing, 2020.
- Wei Wei, T. Andrew Yang, Sadegh Davari, Kewei Sha, and **Jacob, Johanna**. "Toward CAE-CDE 4Y Designation through Curriculum Modernization of a Traditional Computer Science Undergraduate Program". In *Proceedings of the Information Systems and Computing Education Conference (EDSIGCON)*, 2018.
- **Jacob, Johanna**, Wei Wei, Kewei Sha, Sadegh Davari, and T. Andrew Yang. "Is the NICE Cybersecurity Workforce Framework (NCWF) Effective for a Workforce Comprising of Interdisciplinary Majors?". In *Proceedings of the 16th International Conference on Scientific Computing (CSC'18)*, Las Vegas, USA, 2018.
- **Jacob, Johanna**. "Towards a Holistic Interdisciplinary Education in Cybersecurity". Master's Thesis, University of Houston - Clear Lake, 2018.

PRESENTATIONS

- **Jacob, Johanna**. "Rural Cyber Rising: Building Pathways, Expanding Access, and Driving Impact Through a Gamified Intervention". *Virginia Cybersecurity Education Conference, Virginia Tech, July* 2026
- **Jacob, Johanna**. "The Future of Cybersecurity Education: Degrees, Certifications, Micro-Credentials". *Enterprise Security Summit, Kerrville, May* 2026
- **Jacob, Johanna**. "Cybersecurity in the Age of AI: Why This Matters Now for Your Property". *ALP Lodging Conference and Marketplace, March* 2026 (Invited Talk)
- **Jacob, Johanna**. "Engaging Rural Middle and High School Students in Cybersecurity Foundations Through Gamified Interventions: A Case Study". *1st Annual Computer Science Teachers Association Mountains & Plains Virtual Conference, November* 2025
- **Jacob, Johanna**. "An Overview of Possible Inequitable Access to Cybersecurity Educational Resources in the United States". *Women in Cybersecurity Conference, March* 2022
- **Jacob, Johanna**. "Navigating Pathways in Cybersecurity: Competitions, Scholarships, and Career Opportunities for Young Women". *Cyber Career Day, Incarnate Word High School, San Antonio, Fall* 2021 (Invited Talk)

PROFESSIONAL SERVICE

ACM Technical Symposium on Computer Science Education **Fall 2026**
Program Committee Member

SIGCSE Virtual Conference (SIGCSE-V) **Fall 2026**
Program Committee Member

ACM Southeast Conference **Spring 2025, Fall 2025, Fall 2026**
Program Committee Member

Expert Systems with Applications Journal **Spring 2026, Fall 2026**
Reviewer

Computer Science Teachers Association **Spring 2026**
PK-12 Standards Reviewer

Conference for Research on Equitable and Sustained Participation in Engineering, Computing, and Technology (RESPECT) **Fall 2025**
Reviewer

ACM Technical Symposium on Computer Science Education **Fall 2024**
Reviewer

Google Cybersecurity Clinics Fund Proposal Review Panel **Spring 2024**
Reviewer

ACM Technical Symposium on Computer Science Education **Fall 2023**
Program Committee Member

ACM Special Interest Group on Computer Science Education **Fall 2023**
Reviewer

Youth Cybersecurity Workshop and Expo **Spring 2023**
Volunteer, Angelo State University, San Angelo, Texas

Academia Involvement in Community Cybersecurity Conference **Spring 2023**
Volunteer, UTSA San Pedro I, San Antonio, Texas

Consortium for Computing Sciences in Colleges, South Central **Fall 2022**
Subreviewer / External Reviewer

Journal of The Colloquium for Information Systems Security Education **Fall 2022**
Reviewer

Microsoft TEALS Instructor **Spring and Fall 2022**
Cybersecurity Instructor, Seymour High School, Seymour, Texas

Hawaii International Conference on System Sciences **Fall 2021**
Reviewer

Grace Software Inc. **Fall 2024–2026**
Cybersecurity Advisor and Board Member

Newman International Academy **Fall 2026**
Cybersecurity Advisory Team Member

UNIVERSITY SERVICE

Senator, Faculty Senate **Fall 2026**

Member, International Faculty Support Committee **Fall 2026**

Faculty Marshal, Commencement **Spring 2026**

Volunteer, Academic Open House **Fall 2026, Spring 2026**

Mentor, Trinity Cyber Club **Fall 2026**

Member, Curriculum Committee, Trinity Science Alliance **Fall 2026**

MEDIA

Google DC Immersion Experience **Jun 2023**

UTSA Representative for Consortium of Cybersecurity Clinics *Washington D.C*

Represented UTSA as part of Google's official announcement to expand cybersecurity clinics at the Google Headquarters as part of my involvement with the Cybersecurity Clinic at UTSA (Project Xander). The experience involved discussion (on Cybersecurity Education, Training and Awareness for small businesses and non-profits) with Mr. Sundar Pichai, visiting federal offices, and connecting with senators and congressional representatives on leveraging cybersecurity awareness and training for small enterprises.

Press release: https://blog.google/inside-google/message-ceo/commitment-cybersecurity-workforce/

PROFESSIONAL MEMBERSHIPS

Women in Cybersecurity, ACM, IEEE, Texas State Teachers Association, Computer Science Teachers Association, SANS

AWARDS AND SCHOLARSHIPS

SANS Social Hactics Training Scholarship **Summer 2026**

Women in Cybersecurity AI Learning Series Scholarship **Summer 2026**

Women in Cybersecurity Conference Scholarship **2019–2023**

UTSA Graduate School Professional Development Award **Aug 2022**

UTSA Transdisciplinary Team Grand Challenge, First Place **Fall 2022**

Proposal Title: Addressing Ethical Conflicts in Telehealth Data Privacy Regulations

Consortium of Cybersecurity Clinics Scholarship **Jul 2022**

E. Eugene Schultz, Jr., Memorial Scholarship **Jul 2022**

ISSA Alamo Chapter George "Chip" Meadows Memorial Scholarship **Jul 2022**

UTSA Cyber Center for Security and Analytics Travel Award **Mar 2021, Mar 2022**

Women in Cybersecurity Travel Award **Mar 2021**

ISC2 Graduate Cybersecurity Scholarship **Jul 2020**

Other Graduate Travel Awards **Fall 2017–'18**

TECHNICAL SKILLS

Languages: Python, C, Latex, R, HTML/CSS, Javascript
Security Tools/Environment: Kali Linux, Wireshark, Nmap, Burp Suite, Metasploit
Tools/Technologies: IBM SPSS, MATLAB, Tableau, STATA, NVivo
Libraries: Pandas, NumPy, Matplotlib